

Numerical Modeling of Membrane Hydration and Diffusive Water Transport in PEM Water Electrolysis

Selected computational-modeling summary

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Scope note. This portfolio summary presents only the numerical-modeling work described in the project report. Specific geometry dimensions and operating values are intentionally omitted. The implementation was based on publicly available scientific literature and is presented here as a simplified diffusion-only membrane submodel, not as a full PEM electrolyzer model.

Project objective

The goal was to build a simplified numerical model for water diffusion inside a proton-exchange membrane and to examine how different anode-side pore patterns can influence local membrane hydration. The model isolates one transport mechanism - membrane water diffusion - so that the numerical workflow and spatial effects can be studied without the additional coupling present in a full multiphysics electrolyzer model.

What was implemented

- A two-dimensional membrane domain with pore and solid regions prescribed along the anode-side boundary.
- Hydration-dependent membrane water diffusivity, recalculated from the local hydration state during the solution.
- A variable-coefficient steady diffusion equation solved on a finite-difference grid.
- Iterative nonlinear updates until the hydration-related solution converged.
- Post-processing of hydration, local diffusivity and diffusive water-flux fields.
- A comparison of two pore-pattern configurations taken from the public reference literature.

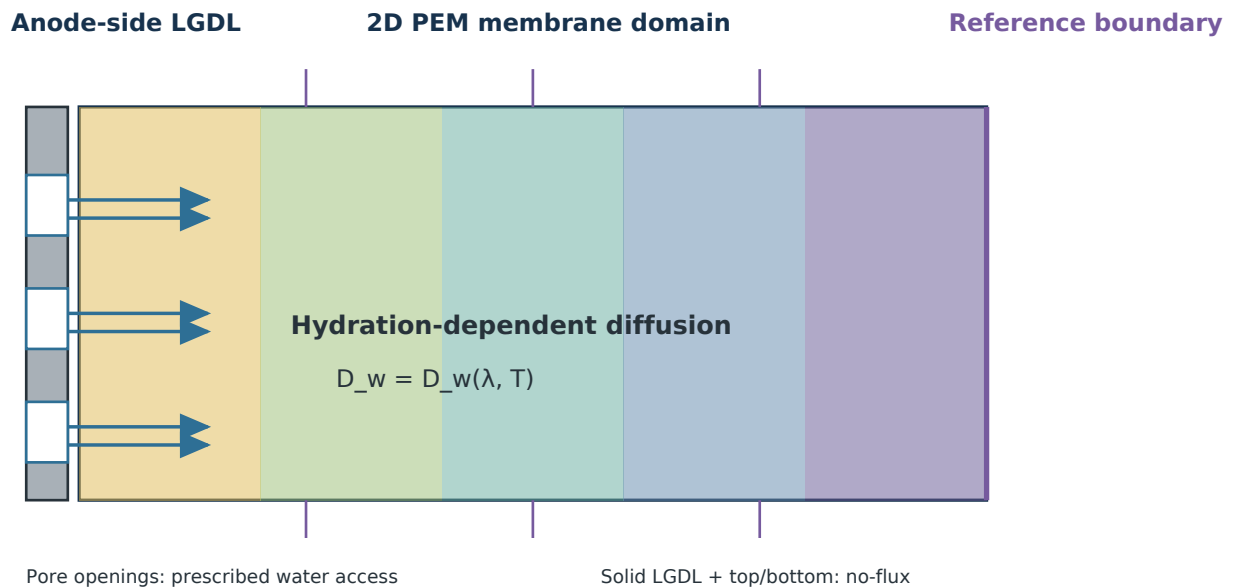
Model scope

The model deliberately excludes electro-osmotic drag, charge transport, electrochemical reaction terms, liquid/gas transport, catalyst-layer transport and flow-channel transport. This makes the project a focused numerical diffusion study rather than a cell-performance model.

1. Conceptual model and governing physics

The membrane was represented as a two-dimensional domain. Pore openings on the anode side act as water-access regions, while the solid parts of the porous transport layer are treated as no-flux boundaries. The opposite membrane surface is assigned a reference hydration state to establish a controlled diffusion gradient.

Conceptual schematic - not to scale



Example numerical result

The figure below shows one of the computed diffusion-only hydration results. Higher hydration develops close to the prescribed water-access regions, while lower hydration appears farther across the membrane and beneath solid boundary regions. The two literature-derived pore patterns create visibly different local field shapes, even though the overall mean hydration remained very similar in this simplified model.

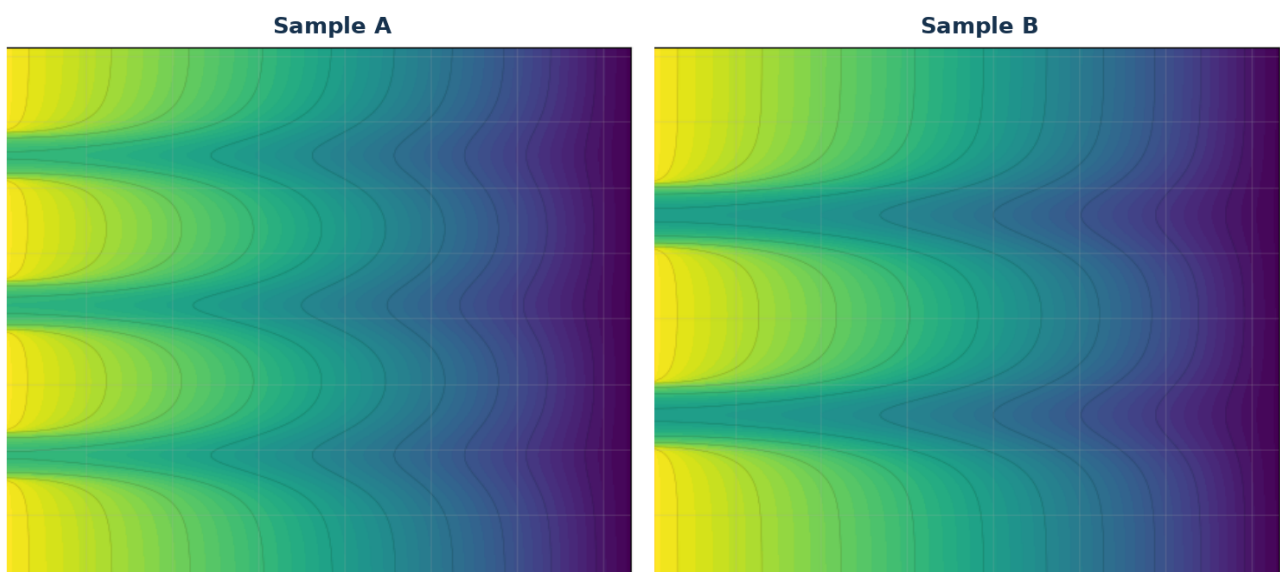
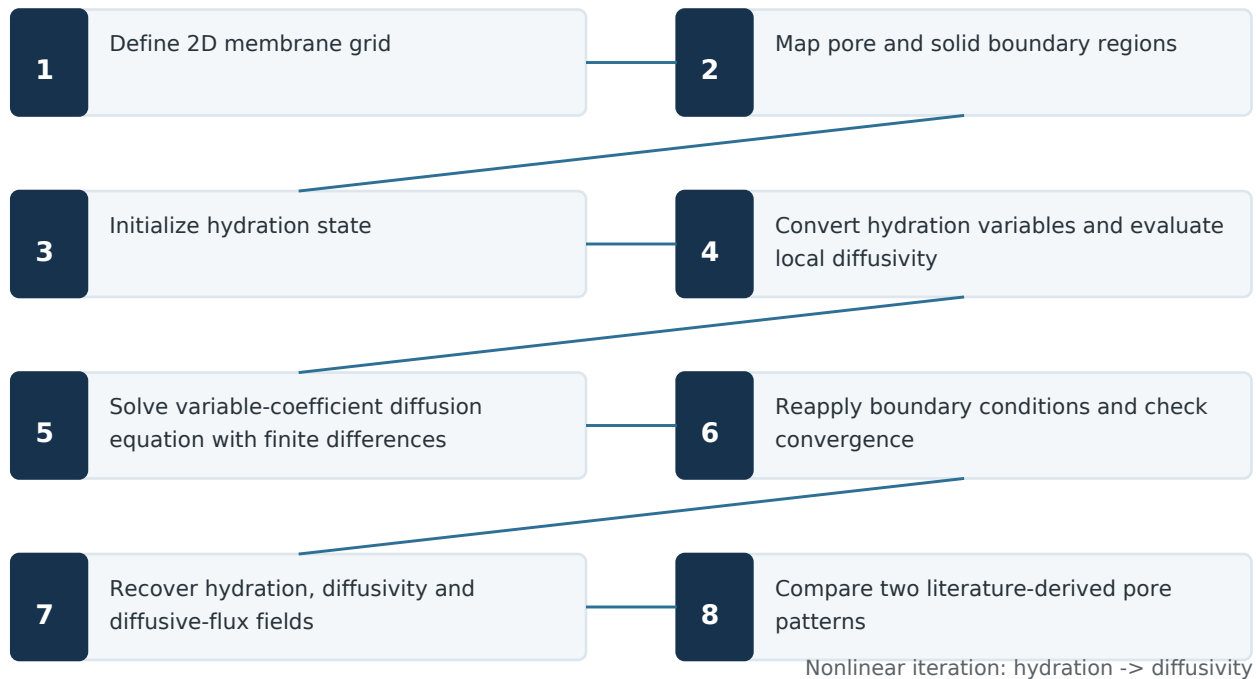


Figure 2. Diffusion-only membrane hydration fields for Sample A and Sample B. Exact coordinate values and geometry dimensions are omitted in this portfolio rendering; the spatial field patterns are taken from the generated model output.

2. Numerical implementation

The diffusion equation was solved on a two-dimensional finite-difference grid. Because hydration affects diffusivity and diffusivity affects the next field update, the calculation was performed iteratively.



Numerical logic

- The current transformed field was converted back to water content and hydration.
- Local membrane diffusivity was recalculated from the updated hydration state.
- The variable-coefficient diffusion equation was updated on the finite-difference grid.
- Boundary conditions were re-applied after each update.
- Iterations continued until the maximum field change became sufficiently small.
- After convergence, the solution was converted back to hydration and the diffusive flux was evaluated from the final gradients.

Why this matters computationally. The project required more than evaluating a closed-form equation: the coefficient field depended on the unknown hydration state, so the solver had to update material properties and the solution repeatedly until a stable field was obtained.

3. Model outputs and interpretation

The numerical workflow produced three field-level outputs. Together they show where water is present in the membrane, how local hydration changes membrane diffusivity, and how the resulting diffusion flux is distributed spatially.

Hydration field Local membrane water content near pore and solid regions. λ	Diffusivity field Spatially varying membrane water diffusivity recalculated from local hydration. $D_w(\lambda, T)$	Diffusive flux field Magnitude and direction of diffusion-driven water transport. $J_{w,diff}$
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Comparison of pore patterns

Two literature-derived pore configurations were evaluated using the same diffusion-only framework. The computed hydration patterns changed locally near the pore openings, but the average membrane hydration differed by only about 0.3% between the two cases in the tested model.

This result was useful because it exposed a limitation of the simplified model: membrane diffusion alone was not sufficient to explain the stronger sample-level behavior reported in the full reference study. The missing mechanisms - especially electro-osmotic drag, current distribution, water supply through the porous layers, reaction effects and gas/liquid transport - are therefore important candidates for a more complete model.

What the project demonstrates

- Translation of a literature-based transport equation into a numerical model.
- Finite-difference discretization of a variable-coefficient diffusion problem.
- Iterative treatment of nonlinear material behavior through hydration-dependent diffusivity.
- Boundary-condition handling for mixed pore, solid and reference regions.
- Field-based post-processing and comparison of alternative geometrical patterns.
- Critical interpretation of model limitations rather than treating simulation output as a full-cell prediction.

4. Reproducibility, limitations and next steps

This project is best viewed as a focused numerical-methods exercise embedded in a PEM electrolysis context. Its value is in the formulation, discretization, iterative solution and interpretation of a physically motivated transport problem.

Current limitations

- Diffusion-only membrane model; no electro-osmotic drag.
- No ionic current distribution or electrochemical reaction coupling.
- No explicit flow-channel, LGDL or catalyst-layer transport.
- No liquid-water saturation or gas-removal model.
- Controlled boundary hydration conditions are modeling assumptions rather than direct experimental boundary conditions.

Natural extensions

- Add electro-osmotic drag and couple water transport to ionic current.
- Extend the domain to include porous transport and catalyst-layer transport.
- Introduce coupled source terms and reaction-dependent boundary conditions.
- Perform grid-convergence and parameter-sensitivity studies for solver verification.
- Compare simplified and coupled-model outputs against published experimental trends.

References

Crank, J. (1975). The Mathematics of Diffusion, 2nd ed., Clarendon Press.

Wrubel, J. A., Kang, Z., Witteman, L., Zhang, F.-Y., Ma, Z., & Bender, G. (2021). Mathematical Modeling of Novel Porous Transport Layer Architectures for Proton Exchange Membrane Electrolysis Cells. International Journal of Hydrogen Energy.

Portfolio note. This summary intentionally omits the specific dimensions and numerical operating values used in the underlying study. It focuses on the computational methods, workflow and interpretation of the modeling work.